

Team Hansel&Gretel at CLEF HIPE-2026 – Short System Description

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Team Hansel&Gretel, affiliated with the Indian Institute of Technology Roorkee, India, focused on the Person–Place Relation Extraction task and submitted three runs exploring progressively larger and less supervised approaches to temporal relation classification in multilingual historical newspaper text. Run 1 builds on a Qwen2.5-3B-Instruct backbone fine-tuned via Low-Rank Adaptation (LoRA; rank 16, $\alpha = 32$, approximately 1.6% trainable parameters) in a Generative Supervised Fine-Tuning (SFT) regime: each training instance is formatted as a chat-template conversation and the model is trained to produce a structured JSON response, with cross-entropy loss masked to output tokens only, so that prediction is framed as controlled generation rather than label classification; gold-annotated documents receive five-times the sample weight of silver-labelled sandbox data, and a Wikidata death-date hard rule post-processes *isAt* predictions at inference time (impresso score: 0.5458). Run 2 replaces the single generalist model with a three-agent architecture: a Historian specialist (temporal and biographical reasoning) and a Geographer specialist (spatial and geographical reasoning), both LoRA fine-tuned on a Qwen2.5-7B-Instruct backbone with three-times class-weighted loss on TRUE/PROBABLE tokens to prevent specialist collapse, independently produce verdicts and free-text reasoning chains that are passed to a fine-tuned Arbiter agent, which resolves disagreements conservatively; three worked few-shot examples per specialist further improve output calibration (impresso score: 0.5788; generalisation score on the out-of-domain test set: 0.6349). Run 3 requires no task-specific model training: it applies Chain-of-Thought few-shot prompting via IBM RITS (GPT-OSS 120B) with dynamic Wikidata knowledge injection, where person birth and death dates together with location descriptions are fetched via SPARQL and prepended as a structured facts block, grounding temporal reasoning in external structured knowledge rather than parametric memory alone (impresso score: 0.6221, rank 13 out of 46 ranked entries).